```
import pandas as pd
import numpy as np
```

```
df=pd.read_csv("/content/Iris.csv")
```

```
df
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['Species']=le.fit_transform(df['Species'])
df
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	0
1	2	4.9	3.0	1.4	0.2	0
2	3	4.7	3.2	1.3	0.2	0
3	4	4.6	3.1	1.5	0.2	0
4	5	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	2
146	147	6.3	2.5	5.0	1.9	2
147	148	6.5	3.0	5.2	2.0	2
148	149	6.2	3.4	5.4	2.3	2
149	150	5.9	3.0	5.1	1.8	2

150 rows × 6 columns

```
y=df['Species']
print("y",y)
```

```
y 0      0
1      0
2      0
3      0
4      0
      ..
145    2
146    2
147    2
148    2
149    2
Name: Species, Length: 150, dtype: int64
```

```
x=df.drop(columns=['Species'])
print("x",x)
```

```
x       Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
0      1            5.1           3.5            1.4           0.2
1      2            4.9           3.0            1.4           0.2
```

```
2      3           4.7           3.2           1.3           0.2
3      4           4.6           3.1           1.5           0.2
4      5           5.0           3.6           1.4           0.2
..   ...           ...           ...           ...           ...
145  146           6.7           3.0           5.2           2.3
146  147           6.3           2.5           5.0           1.9
147  148           6.5           3.0           5.2           2.0
148  149           6.2           3.4           5.4           2.3
149  150           5.9           3.0           5.1           1.8

[150 rows x 5 columns]
```

```
x=np.array(x)
y=np.array(y).reshape(-1,1)
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    x, y, test_size=0.3, random_state=42, stratify=y
)
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
param_grid = {'n_neighbors': range(1, 21)}

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

grid = GridSearchCV(KNeighborsClassifier(), param_grid, cv=cv, scoring='accuracy')
grid.fit(X_train, y_train)

print("Best K:", grid.best_params_)
print("Best CV Accuracy:", grid.best_score_)
```

```
Best K: {'n_neighbors': 2}
Best CV Accuracy: 1.0
```

```
best_knn = grid.best_estimator_

y_pred = best_knn.predict(X_test)
```

```
from sklearn.metrics import accuracy_score, confusion_matrix,f1_score
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Acurracy:",accuracy)
```

```
Acurracy: 1.0
```

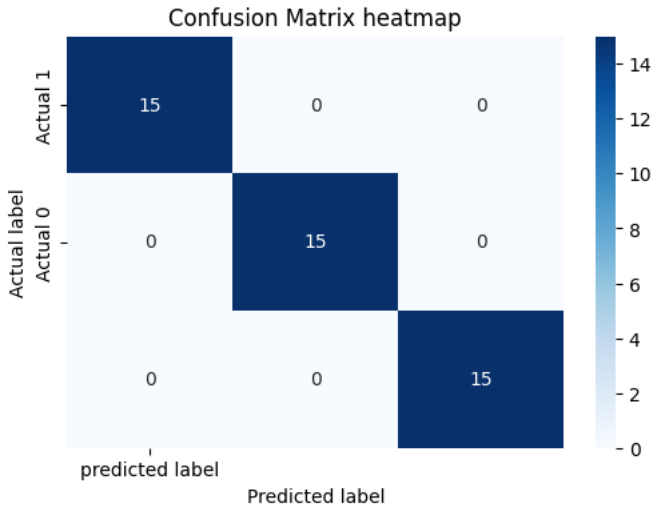
```
f1=f1_score(y_test, y_pred, zero_division=0, average='weighted')
print("F1 Score:",f1)
```

```
F1 Score: 1.0
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm=confusion_matrix(y_test,y_pred)
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['predicted label'], yticklabels=['Actual 1', 'Actual 0'])
plt.xlabel('Predicted label')
plt.ylabel('Actual label')
plt.title('Confusion Matrix heatmap')
plt.show()
```




```
df=pd.read_csv("/content/boston.csv")
```

```
df
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	36.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273.0	21.0	391.99	9.67	22.4
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273.0	21.0	396.90	9.08	20.6
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273.0	21.0	396.90	5.64	23.9
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273.0	21.0	393.45	6.48	22.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273.0	21.0	396.90	7.88	11.9

506 rows × 14 columns

```
# Removed LabelEncoder for 'MEDV' as it's a continuous variable.
# from sklearn.preprocessing import LabelEncoder
# le=LabelEncoder()
# df['MEDV']=le.fit_transform(df['MEDV'])
df
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296.0	15.3	396.90	4.98	137
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	113
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	196
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	193
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	203
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273.0	21.0	391.99	9.67	121
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273.0	21.0	396.90	9.08	104
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273.0	21.0	396.90	5.64	136
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273.0	21.0	393.45	6.48	117
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273.0	21.0	396.90	7.88	26

506 rows × 14 columns

```
y=df['MEDV']
print("y",y)
```

```
y 0      137
1      113
2      196
3      193
4      203
      ...
501    121
502    104
503    136
504    117
505     26
Name: MEDV, Length: 506, dtype: int64
```

```
from sklearn.model_selection import train_test_split

x = df.drop(columns=['MEDV'])
y = df['MEDV']

x=np.array(x)
y=np.array(y).reshape(-1,1)

X_train, X_test, y_train, y_test = train_test_split(
    x, y, test_size=0.3, random_state=42 # Removed stratify=y
)
```

```
from sklearn.neighbors import KNeighborsRegressor
```

```
knn_regressor = KNeighborsRegressor(n_neighbors=5) # You can adjust n_neighbors
knn_regressor.fit(X_train, y_train)
```

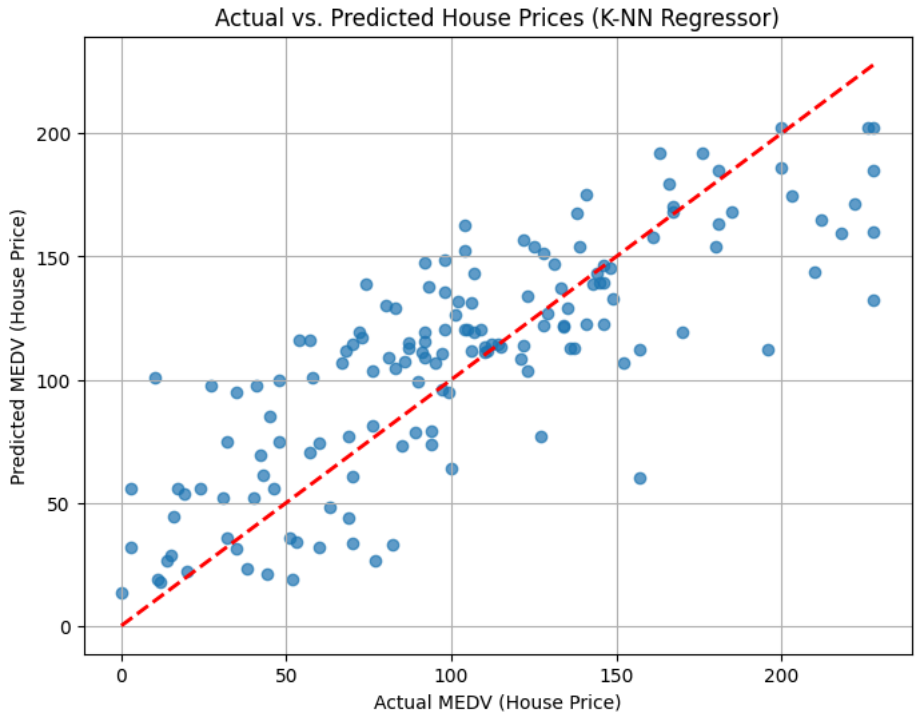
```
▾ KNeighborsRegressor
KNeighborsRegressor()
```

```
y_pred_reg = knn_regressor.predict(X_test)
```

```
from sklearn.model_selection import GridSearchCV, StratifiedKFold, KFold
from sklearn.neighbors import KNeighborsRegressor

param_grid = {'n_neighbors': range(1, 21)}
cv = KFold(n_splits=5, shuffle=True, random_state=42)

grid = GridSearchCV(KNeighborsRegressor(), param_grid, cv=cv, scoring='r2')
grid.fit(X_train, y_train)

print("Best K:", grid.best_params_)
print("Best CV R^2:", grid.best_score_)

y_pred = grid.best_estimator_.predict(X_test)
```

```
Best K: {'n_neighbors': 5}
Best CV R^2: 0.5522555880601253
```

```
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.7)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Actual MEDV (House Price)')
plt.ylabel('Predicted MEDV (House Price)')
plt.title('Actual vs. Predicted House Prices (K-NN Regressor)')
plt.grid(True)
plt.show()
```




```
from sklearn.metrics import mean_squared_error, r2_score
```